Multimodal Large Model + Color Blocks Back In Place（Text Version）

Before running the function, you need to close the App and large programs. For the closing method, refer to [4. Preparation] - [1. Manage APP control services].

1. Function Description

After the program runs, first input the command to remember the current positions of color blocks in the terminal. After the program finishes recording, scramble the positions of the color blocks, then input the command to return the color blocks on the desktop to their original positions. After thinking, the large model will plan an action list, and the program will control the robotic arm to complete the color block return task according to the action functions in the action list.

2. Startup

After entering the Docker container, enter the following commands in the Docker terminal.

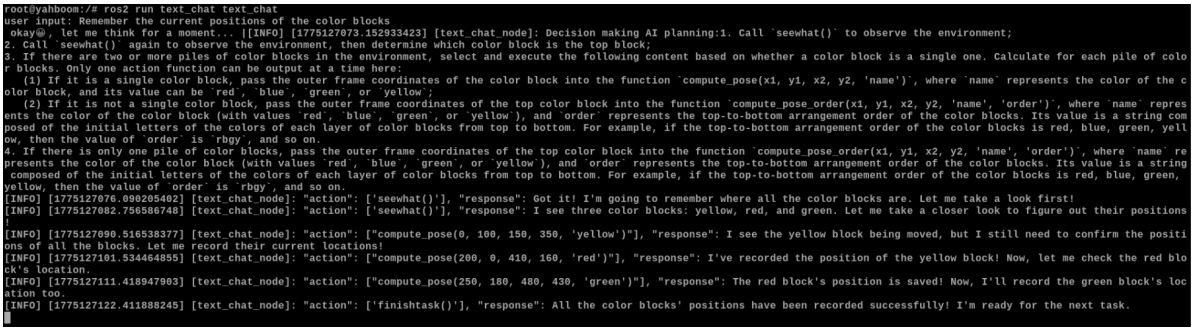
```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

Then open a second terminal and input the following command:

```
ros2 run text_chat text_chat
```

Now input the command to remember the current positions of color blocks in the terminal. You can refer to the following example:

```
Remember the current positions of the color blocks
```



After recording the color block positions, input the end task command in the terminal:

```
End the position recording task
```

Then you can scramble the positions of the color blocks. After scrambling, input the color block return command in the terminal. You can refer to:

```
Return the blocks on the desktop to their original positions
```

```
user input: Return the blocks on the desktop to their original positions
okay@, let me think for a moment... /[INFO] [1775127293.138789408] [text_chat_node]: Decision making AI planning:1. First, call the `color_back_to_orin()` function to control the color blo
cks to return to their original positions.
2. Call the `check_remove(color)` function to check whether the target color block exists in the removal list (where `color` is the color of the target color block, and its value range is `r
ed`, `blue`, `green`, or `yellow`).
3. If the target color block exists in the removal list, perform the following operations:
   (1) Call the `grasp_from_rm_list(color)` function to grasp the target color block from the removal list (the meaning of the `color` parameter is the same as above);
   (2) Pass the color of the target color block as a parameter into the `return_to_orin(color)` function. For example, when the target color block is red, call `return_to_orin(red)` (the val
ue range of the `color` parameter is the same as above).
4. If the target color block does not exist in the removal list, follow the steps below to put the color block back to its original position (only one action function can be output at a time
):
   (1) Call the `seewhat()` function to observe the environment;
   (2) Determine whether there are other color blocks on top of the target color block that needs to be returned to its original position;
   (3) If there are no other color blocks on top of the target color block that needs to be returned to its original position, perform the following steps:
       ① Call the `grasp_obj(x1, y1, x2, y2)` function to grasp the target color block that needs to be returned to its original position (where `(x1, y1, x2, y2)` are the border coordinates
 of the top surface of the target color block that needs to be returned to its original position);
       ② Pass the color of the target color block as a parameter into the `return_to_orin(color)` function (the value range of the `color` parameter is the same as above).
   (4) If there are other color blocks on top of the target color block that needs to be returned to its original position, perform the following steps:
       ① Call the `remove_obj(x1, y1, x2, y2, color)` function to remove the color block (where `(x1, y1, x2, y2)` are the border coordinates of the top surface of this color block, and `col
or` is the color of this color block, with a value range of `red`, `blue`, `green`, or `yellow`);
       ② Call the `seewhat()` function to observe the environment;
       ③ If there are no other color blocks above the target color block that needs to be returned to its original position, call the `grasp_obj(x1, y1, x2, y2)` function to grasp the target
 color block (where `(x1, y1, x2, y2)` are the border coordinates of the top surface of the target color block);
       ④ Pass the color of the target color block as a parameter into the `return_to_orin(color)` function (the value range of the `color` parameter is the same as above).
[INFO] [1775127295.637091377] [text_chat_node]: "action": ['color_back_to_orin()'], "response": I'm going to return all the blocks on the desktop to their original positions. Let me start by
 checking if there are any blocks in the removal list.
[INFO] [1775127298.559595731] [text_chat_node]: "action": ["check_remove('green')"], "response": I need to check if the green block is in the removal list before returning it to its original
 position.
[INFO] [1775127301.237892451] [text_chat_node]: "action": ['seewhat()'], "response": The green block is not in the removal list, so I'll need to check if there are any other blocks on top of
 it before moving it.
[INFO] [1775127308.936447782] [text_chat_node]: "action": ['grasp_obj(50, 60, 210, 220)'], "response": I can see the green block is not covered by any other blocks, so I'll go ahead and gras
p it to return it to its original position.
[INFO] [1775127336.118697487] [text_chat_node]: "action": ["return_to_orin('green')"], "response": I've grasped the green block. Now, I'll return it to its original position.
[INFO] [1775127346.934183907] [text_chat_node]: "action": ["check_remove('red')"], "response": I need to check if the red block is in the removal list before returning it to its original pos
ition.
[INFO] [1775127350.327650876] [text_chat_node]: "action": ['seewhat()'], "response": The red block is not in the removal list, so I'll check if there are any other blocks on top of it before
 moving it.
[INFO] [1775127358.740589262] [text_chat_node]: "action": ['grasp_obj(50, 60, 210, 220)'], "response": I can see the red block is not covered by any other blocks, so I'll go ahead and grasp
it to return it to its original position.
[INFO] [1775127386.690532688] [text_chat_node]: "action": ["return_to_orin('red')"], "response": I've grasped the red block. Now, I'll return it to its original position.
[INFO] [1775127398.498246013] [text_chat_node]: "action": ["check_remove('yellow')"], "response": I need to check if the yellow block is in the removal list before returning it to its origin
al position.
[INFO] [1775127402.594439814] [text_chat_node]: "action": ['seewhat()'], "response": The yellow block is not in the removal list, so I'll check if there are any other blocks on top of it bef
ore moving it.
[INFO] [1775127412.528910495] [text_chat_node]: "action": ['grasp_obj(0, 0, 230, 140)'], "response": I can see the yellow block is not covered by any other blocks, so I'll go ahead and grasp
 it to return it to its original position.
[INFO] [1775127441.938348908] [text_chat_node]: "action": ["return_to_orin('yellow')"], "response": I've grasped the yellow block. Now, I'll return it to its original position.
[INFO] [1775127456.434201391] [text_chat_node]: "action": ['finishtask()'], "response": I have successfully returned all the blocks to their original positions. Everything is back in place!
```

After the buzzer beeps once, the program starts gripping color blocks and returning them to their original positions according to the order recorded in the program. If there are other color blocks above the current color block that needs to be returned, the upper color blocks will be removed first, then the target block will be gripped and returned to its original position.

3. Task Planning

3.1. Record Current Positions of Color Blocks

1. Call `seewhat()` to observe the environment;
2. Call `seewhat()` again to observe the environment, then determine which color block is the top block;
3. If there are two or more piles of color blocks in the environment, select and execute the following content based on whether a color block is a single one. Calculate for each pile of color blocks. Only one action function can be output at a time here:
   (1) If it is a single color block, pass the outer frame coordinates of the color block into the function `compute_pose(x1, y1, x2, y2, 'name')`, where `name` represents the color of the color block, and its value can be `red`, `blue`, `green`, or `yellow`;
   (2) If it is not a single color block, pass the outer frame coordinates of the top color block into the function `compute_pose_order(x1, y1, x2, y2, 'name', 'order')`, where `name` represents the color of the color block (with values `red`, `blue`, `green`, or `yellow`), and `order` represents the top-to-bottom arrangement order of the color blocks. Its value is a string composed of the initial letters of the colors of each layer of color blocks from top to bottom. For example, if the top-to-bottom arrangement order of the color blocks is red, blue, green, yellow, then the value of `order` is `rbgy`, and so on.
4. If there is only one pile of color blocks, pass the outer frame coordinates of the top color block into the function `compute_pose_order(x1, y1, x2, y2, 'name', 'order')`, where `name` represents the color of the color block (with values `red`, `blue`, `green`, or `yellow`), and `order` represents the top-to-bottom arrangement order of the color blocks. Its value is a string composed of the initial letters of the colors of each layer of color blocks from top to bottom. For example, if the top-to-bottom arrangement order of the color blocks is red, blue, green, yellow, then the value of `order` is `rbgy`, and so on.

3.2. Color Blocks Return

1. First, call the `color_back_to_orin()` function to control the color blocks to return to their original positions.
2. Call the `check_remove(color)` function to check whether the target color block exists in the removal list (where `color` is the color of the target color block, and its value range is `red`, `blue`, `green`, or `yellow`).
3. If the target color block exists in the removal list, perform the following operations:
   (1) Call the `grasp_from_rm_list(color)` function to grasp the target color block from the removal list (the meaning of the `color` parameter is the same as above);
   (2) Pass the color of the target color block as a parameter into the `return_to_orin(color)` function. For example, when the target color block is red, call `return_to_orin(red)` (the value range of the `color` parameter is the same as above).
4. If the target color block does not exist in the removal list, follow the steps below to put the color block back to its original position (only one action function can be output at a time):
   (1) Call the `seewhat()` function to observe the environment;
   (2) Determine whether there are other color blocks on top of the target color block that needs to be returned to its original position;
   (3) If there are no other color blocks on top of the target color block that needs to be returned to its original position, perform the following steps:
   ①Call the `grasp_obj(x1, y1, x2, y2)` function to grasp the target color block that needs to be returned to its original position (where `(x1, y1, x2, y2)` are the border coordinates of the top surface of the target color block that needs to be returned to its original position);
   ②Pass the color of the target color block as a parameter into the `return_to_orin(color)` function (the value range of the `color` parameter is the same as above).
   (4) If there are other color blocks on top of the target color block that needs to be returned to its original position, perform the following steps:
   ①Call the `remove_obj(x1, y1, x2, y2, color)` function to remove the color block (where `(x1, y1, x2, y2)` are the border coordinates of the top surface of this color block, and `color` is the color of this color block, with a value range of `red`, `blue`, `green`, or `yellow`);
   ②Call the `seewhat()` function to observe the environment;
   ③If there are no other color blocks above the target color block that needs to be returned to its original position, call the `grasp_obj(x1, y1, x2, y2)` function to grasp the target color block (where `(x1, y1, x2, y2)` are the border coordinates of the top surface of the target color block);
   ④Pass the color of the target color block as a parameter into the `return_to_orin(color)` function (the value range of the `color` parameter is the same as above).

4. Core Code Analysis

4.1. compute_pose_order Record Stacked Color Block Position

Source code path: `LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
#Parameter description
#x1,y1,x2,y2: Top outer border coordinates of the top color block, x1,y1
represent
#name: Color of the top color block
```

```
#order: Represents the stacking order list from top to bottom, for example if the stacked colors from top to bottom are red and yellow, then the passed order is ry
def compute_pose_order(self,x1,y1,x2,y2,name,order):
    #Get current color block parameters
    cur_name = name.strip("'\"")
    color_order = order.strip("'\"")
    #Create a list to store the stacking order of color blocks from top to bottom
    order_list = list(color_order[1:])
    self.get_logger().info(f"order_list:{order_list}")
    self.cur_name = cur_name
    self.cur_pose[self.cur_name] = []
    #Start position calculation program
    cmd1 = "ros2 run largemodel_arm Record_pose"
    proc = subprocess.Popen(
        [
            "gnome-terminal",
            "--title=compute_pose",
            "--",
            "bash",
            "-c",
            f"{cmd1}; exec bash",
        ]
    )
    self.pid_num = proc.pid
    self.get_logger().info(f"self.pid_num:{self.pid_num}")
    time.sleep(5.0)
    x1 = int(x1)
    y1 = int(y1)
    x2 = int(x2)
    y2 = int(y2)
    self.object_position_pub.publish(Int16MultiArray(data=[x1, y1, x2, y2]))

    while not self.compute_pose_future.done():
        if self.interrupt_flag:
            self.check_close_compute_pose()
            Arm.Arm_serial_servo_write6(90,150,12,20,90,30,1000)
            self.stop()
            return
        time.sleep(0.1)

    result = self.compute_pose_future.result()
    self.check_close_compute_pose()
    if not self.interrupt_flag:
        if result.data == "compute_pose_done":
            for i in range(0,len(order_list)):
                #Here we consider that xy directions are consistent with the top block, only height changes, so just need to change z value, decreasing by 2.8cm each time, approximately the height of one color block
                x = self.cur_pose[cur_name][0]
                y = self.cur_pose[cur_name][1]
                z = self.cur_pose[cur_name][2] - (i+1)*0.028
                if order_list[i] == 'r' or  order_list[i] == 'R':
                    self.cur_pose['red'] = []
                    self.cur_pose['red'].append(x)
                    self.cur_pose['red'].append(y)
                    self.cur_pose['red'].append(z)
```

```
                elif order_list[i] == 'b' or  order_list[i] == 'B':
                    self.cur_pose['blue'] = []
                    self.cur_pose['blue'].append(x)
                    self.cur_pose['blue'].append(y)
                    self.cur_pose['blue'].append(z)
                elif order_list[i] == 'y' or  order_list[i] == 'Y':
                    self.cur_pose['yellow'] = []
                    self.cur_pose['yellow'].append(x)
                    self.cur_pose['yellow'].append(y)
                    self.cur_pose['yellow'].append(z)
                elif order_list[i] == 'g' or  order_list[i] == 'G':
                    self.cur_pose['green'] = []
                    self.cur_pose['green'].append(x)
                    self.cur_pose['green'].append(y)
                    self.cur_pose['green'].append(z)

            self.get_logger().info(f"self.cur_pose:{self.cur_pose}")
            #Feedback to large model that recording stacked color block
positions is complete
            self.action_status_pub("compute_pose_order_done", x1=x1, y1=y1,
x2=x2, y2=y2,name=name,order=order)
        else:
            self.action_status_pub("compute_pose_failed", x1=x1, y1=y1, x2=x2,
y2=y2)
```

4.2. color_back_to_orin Publish Color Block Return Order

Source code path: `LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
def color_back_to_orin(self):
    self.back_list = True
    #Buzzer beeps once
    Arm.Arm_Buzzer_On()
    time.sleep(0.5)
    Arm.Arm_Buzzer_Off()
    #Traverse the list storing color block positions in reverse order, feedback
to large model the current color block return instruction
    for key in reversed(self.cur_pose.keys()):
        if key == 'red':
            self.action_status_pub("send_red_back_to_orin")
        elif  key == 'green':
            self.action_status_pub("send_green_back_to_orin")
        elif key == 'blue':
            self.action_status_pub("send_blue_back_to_orin")
        elif key == 'yellow':
            self.action_status_pub("send_yellow_back_to_orin")
        while not self.return_done:
            time.sleep(0.1)
        self.return_done = False
    self.action_status_pub("return_to_orin_done")
    self.back_list = False
"send_red_back_to_orin": "Robot feedback: Return the red block to its original
position",
"send_green_back_to_orin": "Robot feedback: Return the green block to its
original position",
"send_blue_back_to_orin": "Robot feedback: Return the blue block to its original
position",
```

```
"send_yellow_back_to_orin": "Robot feedback: Return the yellow block to its
original position"
```

4.3. check_remove(color) Check if Current Target Color Block Exists in Removal List

Source code path: `LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
def check_remove(self,color):
    #Get current color block color
    check_color = color.strip("'\"")
    #If current color block color can be found in removal list, then feedback to
large model that current color block can be found in removal list
    if check_color in self.cur_rm_pose:
        self.action_status_pub("color is in the rm_list")
    else:
        self.action_status_pub("color is not in the rm_list")

"color is in the rm_list" : "Color block exists in removal list, grip this color
block from removal list",
"color is not in the rm_list" : "Color block does not exist in removal list,
determine if there are other objects on top of this color block",
```

4.4. grasp_from_rm_list Grip Color Block from Removal List

Source code path: `LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
def grasp_from_rm_list(self,color):
    #Get current color block color
    tar_color = color.strip("'\"")
    #Find corresponding robotic arm posture value in removal dictionary based on
current color
    tar_joints = self.cur_rm_pose.get(tar_color)
    Arm.Arm_serial_servo_write6(tar_joints[0], tar_joints[1], tar_joints[2],
tar_joints[3], tar_joints[4], tar_joints[5],2000)
    time.sleep(2.0)
    Arm.Arm_serial_servo_write(6, 140, 1000)
    time.sleep(2.0)
    Arm.Arm_serial_servo_write6(90,120,10,10,90,140,2000)
    time.sleep(2.0)
    #Feedback to large model that gripping from removal list is complete
    self.action_status_pub("grasp_from_rm_list_done")
```